

Chapter 11 – Wi-Fi Attacks, Handshake Capture, Evil Twin Attack & Aircrack-ng

Introduction to Wi-Fi Attacks

A **Wi-Fi Attack** is a cyber attack where an attacker targets a wireless network to:

- Gain unauthorized access 
- Capture user data 
- Crack Wi-Fi passwords 
- Monitor traffic 
- Disconnect users 
- Create fake Wi-Fi networks 

These attacks mostly target weaknesses in:

- WPA/WPA2 security
- Weak passwords
- WPS
- User trust
- Wireless protocols

Basic Requirement for Wi-Fi Attacks

For most wireless attacks in Kali Linux, you need:

External USB Wi-Fi Adapter (Dongle)

A normal laptop Wi-Fi card usually does **not support monitor mode and packet injection** properly.

So attackers use a separate USB Wi-Fi adapter.

◆ Important Features Required:

- Monitor Mode Support
- Packet Injection Support

◆ Popular Adapters:

- Alfa AWUS036NHA
- TP-Link TL-WN722N (v1)



What is Monitor Mode?

Normally, your Wi-Fi card only listens to packets meant for your device.

But in **Monitor Mode**, the adapter captures **all wireless packets** around it.

This allows:

- Packet sniffing
- Handshake capture
- Deauthentication attacks
- Wi-Fi monitoring



What is airmmon-ng?

airmon-ng is a tool from the Aircrack-ng suite used to enable or disable monitor mode.



Check Wireless Interfaces

airmon-ng

Shows available wireless interfaces like:

wlan0

wlan1

Start Monitor Mode

```
airmon-ng start wlan0
```

After enabling monitor mode:

```
wlan0mon
```

will be created.

Stop Monitor Mode

```
airmon-ng stop wlan0mon
```

Kill Conflicting Processes

Some services interfere with monitor mode.

```
airmon-ng check kill
```

Kills:

- NetworkManager
 - wpa_supplicant
 - dhclient
-

How Wi-Fi Authentication Works

When a device connects to Wi-Fi, it sends the password through a process called:

WPA/WPA2 Handshake

The handshake checks:

- ✓ Is the password correct?
- ✓ Can the device join the network?

During this process:

- Password is not sent directly

- It is converted into a hash

Attackers capture this handshake and try to crack it offline using a wordlist.



What is a Handshake Capture?

A **handshake file** contains encrypted authentication data exchanged between:

- Router
- Client device

The password inside is hashed.

Attackers later compare hash values using wordlists.



Real Life Example

Imagine:

- Wi-Fi password = key 
- Handshake = security guard checking the key

The attacker secretly records this checking process and later tries many duplicate keys until one matches.



Important Point About Handshake

Handshake only happens when:

- A user connects
- Reconnects
- Enters password again

But what if users are already connected?

Then attackers force reconnection using:



Deauthentication Attack

What is a Deauthentication Attack?

The attacker sends fake deauthentication packets to connected users.

Result:

- Devices disconnect from Wi-Fi
- Devices automatically reconnect
- Handshake happens again
- Kali captures handshake

Real Life Example

Suppose someone enters a room using a password-protected door.

You force them outside.

When they re-enter, you secretly record the password verification process.

Step-by-Step Wi-Fi Handshake Capture

◆ Step 1 – Check Wi-Fi Adapter

`iwconfig`

Look for:

`wlan0`

◆ Step 2 – Enable Monitor Mode

`airmon-ng start wlan0`

Now interface becomes:

`wlan0mon`

◆ Step 3 – Scan Nearby Wi-Fi Networks

airodump-ng wlan0mon

This shows nearby Wi-Fi networks.



Understanding Airodump-ng Columns

Column Meaning

BSSID Router MAC Address

PWR Signal strength

Beacons Router broadcast packets

#Data Captured packets

CH Channel number

MB Maximum speed

ENC Encryption type

CIPHER Encryption algorithm

AUTH Authentication method

ESSID Wi-Fi name



Important Terms

◆ BSSID

MAC address of router.

Example:

00:17:7C:C8:4E:EA

◆ PWR

Signal strength.

Closer to 0 = stronger signal.

Example:

-48 stronger than -80

◆ ENC

Encryption type:

- WPA2
- WPA3
- OPN

⚠ WPA3 is much harder to attack.

◆ Step 4 – Capture Handshake

```
airodump-ng --bssid <target_bssid> -c <channel> -w capture wlan0mon
```

Example:

```
airodump-ng --bssid 00:11:22:33:44:55 -c 6 -w capture wlan0mon
```

This saves packets into:

capture-01.cap

◆ Step 5 – Force Handshake Using Deauth Attack

Open another terminal:

```
aireplay-ng --deauth 10 -a <bssid> -c <client_mac> wlan0mon
```

Example:

```
aireplay-ng --deauth 10 -a 00:11:22:33:44:55 -c AA:BB:CC:DD:EE:FF wlan0mon
```

What Happens Here?

- Connected device disconnects
- Auto reconnect triggers
- WPA handshake occurs again
- Kali captures it

If handshake captured successfully, top-right corner shows:

WPA Handshake: <BSSID>

What is aircrack-ng?

aircrack-ng is used to crack WPA/WPA2 passwords using:

- Captured handshake
 - Wordlist
-

Crack Wi-Fi Password

aircrack-ng capture-01.cap -w /usr/share/wordlists/rockyou.txt

Explanation

Part	Meaning
capture-01.cap	Handshake capture file
-w	Wordlist option
rockyou.txt	Password dictionary

How Password Cracking Works

The tool:

1. Takes a password from wordlist
 2. Converts it into hash
 3. Compares with captured hash
 4. If matched → Password found
-

Important Note

Password cracking only works if:

- ✓ Password exists inside the wordlist

If not present:

- ✗ Password cannot be cracked
-



What is Packet Sniffing?

Packet sniffing means capturing and analyzing network packets.

Tools:

- Wireshark
- Bettercap
- tcpdump

Attackers can view:

- Source IP
 - Destination IP
 - Protocols
 - MAC addresses
 - Data packets
-



Types of Wireless Attacks

Attack	Purpose
Deauthentication	Disconnect users
Evil Twin	Fake Wi-Fi
Packet Sniffing	Capture traffic
Password Cracking	Crack Wi-Fi password
Wi-Fi Jamming	Disrupt network
MITM	Intercept communication



Important Wireless Attack Tools

Tool	Use
Aircrack-ng	WPA/WPA2 cracking
Reaver	WPS attacks
Fluxion	Evil Twin + phishing
Bettercap	MITM attacks
Kismet	Wireless IDS
MDK4	Wi-Fi jamming
Wifite	Automated attacks



What is an Evil Twin Attack?

An Evil Twin Attack creates a fake Wi-Fi access point with the same name as the real Wi-Fi.

Victims unknowingly connect to it.

Then attacker can:

- Steal passwords
- Capture traffic
- Launch phishing attacks
- Perform MITM attacks



Real Life Example

Suppose:

- Original Wi-Fi = "Home_WiFi"
- Attacker creates fake "Home_WiFi"

Users accidentally connect to fake one because it looks identical.



Evil Twin Attack Using Fluxion & Aircgeddon

Popular tools:

- Fluxion
- Aircgeddon

These tools automate:

- Deauthentication
- Fake AP creation
- Fake login page hosting
- Password verification

How Fluxion Attack Works

1. Attacker scans nearby Wi-Fi
2. Selects target Wi-Fi
3. Disconnects users using deauth packets
4. Creates fake Wi-Fi with same name
5. Hosts fake login page
6. User enters password
7. Password is captured

Fake Firmware Update Trick

Fluxion may display:

“Router firmware updated. Please enter Wi-Fi password.”

This page is completely fake.

Victim trusts it and enters password.

Important Point

Unlike aircrack-ng:

- ✓ Fluxion does NOT need a wordlist.

Because:

- User directly types the real password.

Install Fluxion

```
sudo apt update
```

```
sudo apt install -y git aircrack-ng isc-dhcp-server hostapd lighttpd php
```

```
git clone https://github.com/FluxionNetwork/fluxion.git
```

```
cd fluxion
```

```
sudo ./fluxion.sh
```



Step-by-Step Fluxion Attack

◆ Step 1 – Start Fluxion

```
sudo ./fluxion.sh
```

◆ Step 2 – Select Language

Choose preferred language.

◆ Step 3 – Select Interface

Example:

```
wlan0
```

◆ Step 4 – Enable Monitor Mode

Fluxion asks automatically.

Choose:

```
Yes
```

◆ Step 5 – Scan Nearby Networks

Tool scans nearby Wi-Fi networks.

◆ Step 6 – Select Target Wi-Fi

Choose:

- Target SSID
 - Channel
 - BSSID
-

◆ Step 7 – Select Attack Method

Usually:

Captive Portal

◆ Step 8 – Capture Handshake

Fluxion captures WPA handshake.

◆ Step 9 – Create Fake AP

Fake Wi-Fi with same SSID becomes active.

◆ Step 10 – Deauthenticate Users

Connected users disconnect from real AP.

They reconnect to fake AP.

◆ Step 11 – Victim Enters Password

Captive portal asks:

“Enter Wi-Fi password.”

Victim enters it.

◆ Step 12 – Password Verification

Fluxion checks password using captured handshake.

If correct:

[+] Correct password found

● Stop Fluxion & Restore Network

```
sudo service NetworkManager restart
```

```
sudo airmon-ng stop wlan0mon
```

🔍 What is Reaver? (WPS Attack)

Reaver attacks routers with WPS enabled.

🔍 Check WPS Routers

```
wash -i wlan0mon
```

🚀 Start Reaver Attack

```
reaver -i wlan0mon -b <bssid> -vv
```

📘 What is Wi-Fi Jamming?

Wi-Fi jamming disrupts wireless communication using:

- Noise packets
 - Repeated deauth packets
-

MDK4 Jamming Example

```
mdk4 wlan0mon d
```

Attack all networks.

Attack Specific Targets

```
mdk4 wlan0mon a -i blacklist.txt
```

blacklist.txt Example

```
00:11:22:33:44:55  
AA:BB:CC:DD:EE:FF
```

Summary of Aircrack-ng Tools

Tool	Purpose
airmon-ng	Enable monitor mode
airodump-ng	Capture packets
aireplay-ng	Inject packets
aircrack-ng	Crack passwords

Prevention Against Wi-Fi Attacks

Protection	Why Important
Use WPA3	Strong security
Disable WPS	Prevent brute force
Strong Passwords	Hard to crack
Update Firmware	Fix vulnerabilities

Protection	Why Important
Use VPN	Encrypt traffic
Monitor Traffic	Detect attacks
MAC Filtering	Restrict devices

⚠ Legal Warning

These tools should only be used for:

- ✓ Ethical hacking
- ✓ Labs
- ✓ Penetration testing
- ✓ CTF practice

Unauthorized Wi-Fi attacks are illegal and unethical.